MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

1/ TYPE DOD-010

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Passive TR-limiter, bandpass, frequency range (see note 1)

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	tp	Pressurization	Temperature		Alt
					Operating	Nonoperating	
Unit:	kw	---	µs	psia	°C	°C	ft
Maximum:	35	0.035	3.0	40	+70	+75	50,000
Minimum:	0	---	---	---	0	-40	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Flange: Mates with UG-1729/U of MIL-F-3922/52
Flange(finish: See figure 1

TEST CONDITIONS:

Parameter:	Incident power	(A) Du	(A) tp	F	Frequency	
					F	MHz
Unit:	kw	---	µs	MHz	P3	±5
Tolerance:	±10%	max	±0.10	±1%	P7	±5
Test condition 1:	6	0.001	2.0	P11	P11	±5
Test condition 2:	35	0.001	2.0	P11	P15	±5
					P18	±5

GENERAL:

(A) First article sample inspection - Required (see note 2)
Service life guarantee - See notes 3 and 4
Shelf life - See note 4

1/ Replaces electron tube described by NOSC Drawing No. 2667877 (Micro Dynamics Inc. P/N MD 80C8)

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
(A)	First article sample inspection							
4021	Degradation due to vibration	-	-		---	---	---	---
1042	Shock, specified pulse	5	-	Condition A, except 40 G	---	---	---	---
1027	Temperature cycling life test	-	-	10 cycles (min)	---	---	---	---
1011	Humidity	-	-		---	---	---	---
	Quality conformance inspection, part 1	6	-					
4473	Low-level VSWR	7	-	F = P3	σ	---	1.35	---
				F = P7	σ	---	1.35	---
				F = P11	σ	---	1.35	---
				F = P15	σ	---	1.35	---
				F = P18	σ	---	1.35	---
4416	Insertion loss	7	-	F = P3	Li	---	0.5	dB
				F = P7	Li	---	0.5	dB
				F = P11	Li	---	0.5	dB
				F = P15	Li	---	0.5	dB
				F = P18	Li	---	0.5	dB
(A)4452	Flat-leakage power	10	1		pf	---	100	mw
(A)4452	Spike-leakage energy	10	1		Ws	---	0.1	erg
(A)4496	Firing power	10	1		Po/Du	---	150	mw
1027	Temperature cycling (nonoperating)	8	-	1 cycle; TA = -62° to +75°C	---	---	---	---
	Quality conformance inspection, part 2							
(A)4471	Recovery time	6, 9	2	TA = 0°C	t	---	2	µs
4101	Dielectric material strain	-	-		---	---	---	---
	Quality conformance inspection, part 3							
(A)---	Life test	3, 4, 10	2	Group D; TA = 70°C	t	2,000	---	hrs
---	Life-test end points:							
4471	Recovery time	9	2		t	---	3	µs
4452	Flat-leakage power	-	1		pf	---	130	mw
4452	Spike-leakage energy	-	1		Ws	---	0.2	erg
4416	Insertion loss	7	-		Li	---	0.5	dB

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NOTES:

1. The frequency range of the tube specified herein and the frequencies of codes P3 through P18 are as specified in Naval Ordance Systems Command (NOSC) Drawing No. OD15440.

(A) 2. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previous procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

3. Life tests shall not be required after the TUT have satisfied all requirements of this specification for first article approval. However, a service-life guarantee of 2,000 hours minimum shall be required in accordance with the requirements of MIL-E-1. Life for purpose of the service-life guarantee shall be defined as the total rf operating time.

4. The tubes tested and procured in accordance with this tube specification sheet shall be capable of shelf-life for a period of 18 months after date of shipment from the manufacturer's plant. Tubes placed in operation during this shelf-life period shall meet all requirements of this tube specification sheet. The sum of the operating and shelf life shall be not less than 21 months. Tubes found to be defective within the shelf-life period shall be replaced by the manufacturer at no cost to the Government or its contractors.

5. The TUT shall be capable of satisfying all requirements of this TSS without damage or failure before, during and after being subjected to the special test when clamped by its input flange and a weight of 2.5 pounds suspended on the output flange, whose center of gravity 4 ±0.25 inches (101.6 ±6.35 mm) from the output flange surface.

6. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

7. A swept frequency signal may be used for this test.

8. After the completion of this test, the TUT shall satisfy all requirements of method 4452, spike-leakage energy, as specified in quality conformance inspection, part 1.

9. The recovery time shall be measured at the point where the signal level has recovered to within 3.0 dB of the full signal level.

(A) 10. The tube must be capable of meeting this requirement when used in a system operating under the conditions as specified in maximum rating section.

Custodian:
Navy - EC

Review activity:
DSA - ES

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N113)

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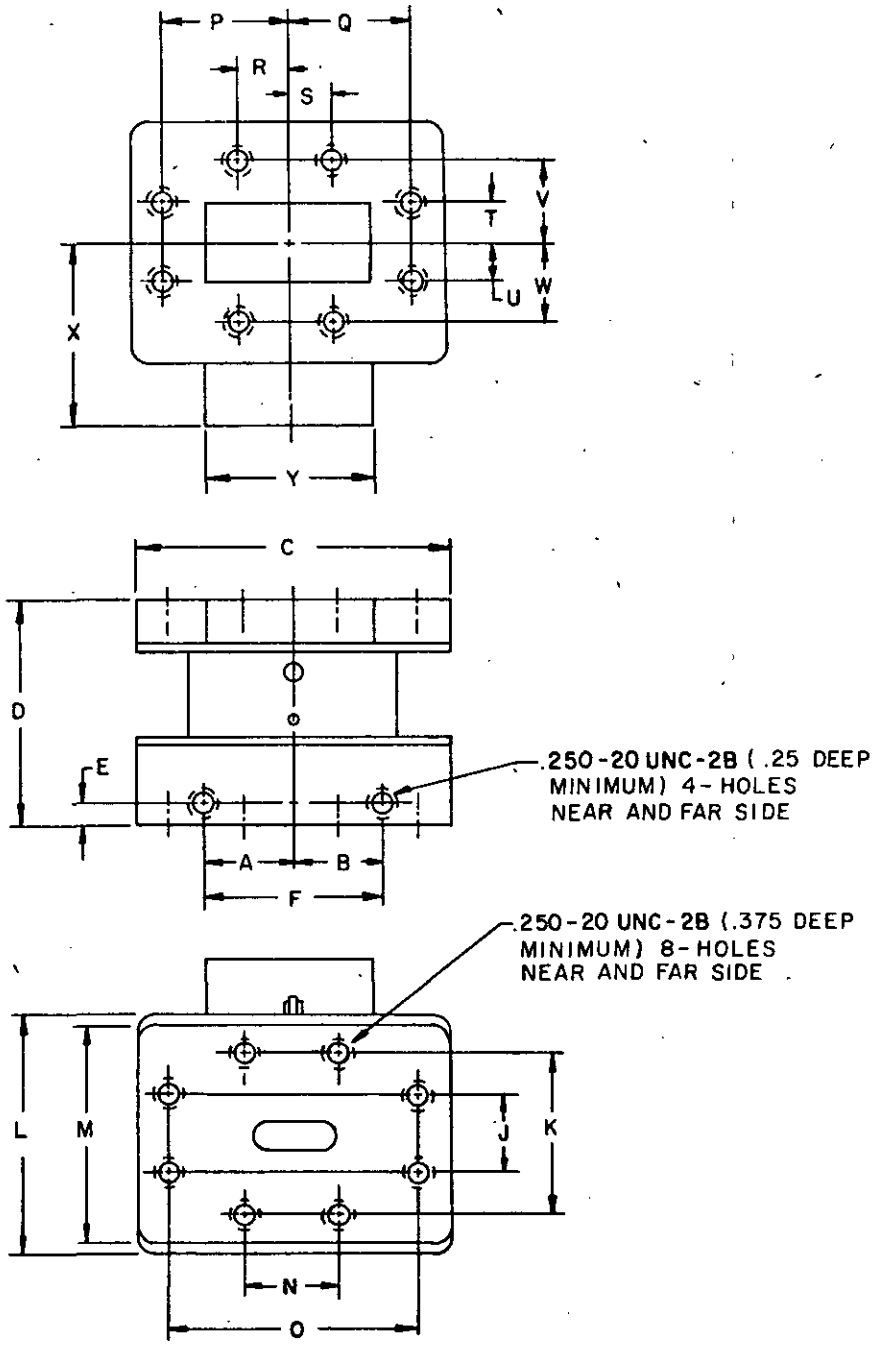


FIGURE 1. Outline drawing of electron tube type DQD-010.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Ⓐ	First article sample inspection		
A	.990(25.15)	1.010(25.65)	a
B	.990(25.15)	1.010(25.65)	a
C	3.200(81.28)	3.800(96.52)	
E	.270 (6.86)	.310 (7.87)	
F	1.980(50.29)	2.020(51.31)	
Quality conformance inspection, part 1			6
Ⓐ D	3.800(96.52)	3.890(98.81)	
P	1.412(35.86)	1.416(35.97)	a
Q	1.412(35.86)	1.416(35.97)	a
R	.561(14.25)	.565(14.35)	a
S	.561(14.25)	.565(14.35)	a
T	.435(11.05)	.439(11.15)	a
U	.435(11.05)	.439(11.15)	a
V	.912(23.16)	.916(23.27)	a
W	.912(23.16)	.916(23.27)	a
X		2.000(50.80)	
Y		2.000(50.80)	
Quality conformance inspection, part 2			
J	.871(22.12)	.879(22.33)	
K	1.824(46.33)	1.832(46.53)	
L	2.850(72.39)	2.910(73.91)	
M	2.470(62.74)	2.530(64.26)	
N	1.122(28.50)	1.130(28.70)	
O	2.824(71.73)	2.832(71.93)	

NOTES:
a. The following sets of dimensions shall be equal within .005(.13 mm): A and B, P and Q, R and S, T and U, and V and W.
b. The input and output flange shall be chemically treated in accordance with MIL-C-5541 (iridite), grade C, class 1. The center portion tube flanges shall be cadmium plated with chromate film as specified by NAVORD OD15444, type 24.

FIGURE 1. Outline drawing of electron tube type DOD-010 - Continued.

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